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Stimulation of Ovarian Maturation in Freshwater Crab, *Oziotelphusa senex senex* Fabricius by Leucine Enkephalin

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Abstract: Enkephalins are the peptide neurotransmitters, also play a role in the crustacean reproduction. In the present study administration of leucine enkephalin at the dose of 10^{-8} mole/crab on 1st, 7th, 14th and 21st day of experiment resulted in significant increase in ovarian growth of freshwater crab *Oziotelphusa senex senex* Fabricius. These results provide evidence to support the hypothesis that enkephalins act as gonadotropins in crustaceans.

Key words: Ovarian Index • Leucine Enkephalin • Histological Studies • Crab

INTRODUCTION

Since the discovery of opioid peptides in crustaceans [1], many studies were conducted to demonstrate the occurrence of these peptides in several crustaceans [2]. Even though there is a large body of immunological evidence for the presence of opioid peptides in crustaceans, relatively few data have been provided concerning their likely physiological activities. Among the described functions of enkephalins are excitatory action for locomotor activity [3], an increased escape response of *Gecarcinus lateralis*, dark adaptation of distal retinal pigment of *Uca pugnator* [4], regulation of hemolymph sugar level in crabs, *Oziotelphusa senex senex* [5], *Scylla serrata* [6] and in prawns *Penaeus indicus* and *Metapenaeus monoceros* [7] and also they act as neurotransmitters.

Methionine enkephalin stimulated release of a red-pigment concentrating hormone and a black-pigment concentrating hormone isolated from eyestalk tissues of *Uca pugnator* [4]. In the same crab, Kulkarni and Fingerman [8] found that the enkephalin stimulated release of the distal retinal pigment dark-adapting hormone [8]. In *Carcinus maenas*, leucine enkephalin inhibited release of crustacean hyperglycemic hormone

[9]. However there are no reports on the effects of leucine enkephalin on the mobilization of other neuro-hormones of the sinus gland.

In view of this an elaborate programme was initiated in our laboratory to evaluate the physiological role(s) of enkephalins in crustaceans. We have found a neurotransmitter role for methionine enkephalin in regulating hemolymph sugar level in fresh water crab, *Oziotelphusa senex senex* [5]. Methionine enkephalin induces hyperglycemia through eyestalk hormone in the estuarine crab, *Scylla serrata* [6] and in prawns *Penaeus indicus* and *Metapenaeus monoceros* [7]. The present report is a part of the above mentioned programme and determines the role of leucine enkephalin in regulating the ovarian maturation in fresh water crab *Oziotelphusa senex senex*.

MATERIALS AND METHODS

Intact, inter molt (Stage C4) adult female *Oziotelphusa senex* (body weight 30 ± 2 g) were collected from the rice fields and irrigation canals around Tirupati, India. They were acclimatized to the laboratory conditions (temperature $27 \pm 1^\circ\text{C}$; Relative Humidity 75% and a light period 12 h) for 7 days in large glass aquaria partially filled

with tap water. The crabs were fed *ad libitum* once in every two days with sheep meat and transferred to fresh medium every day.

Leucine enkephalin was purchased from Sigma Chemical Company and dissolved in crustacean saline [10]. In the present study, 75 crabs having a white (immature) ovary were selected and divided into three groups each consisting of 25 crabs. The crabs in the first group served as initial control did not receive any treatment and was sacrificed on the first day of experiment. The crabs in the second group served as concurrent control and was treated the same as the experimental group but received injections of crustacean saline only [10]. The crabs in the third group received injections of 10^{-8} mol/crab of leucine enkephalin in 10 μ l volume per injection. The injections were given on the 1st, 7th, 14th and 21st day through the arthrodial membrane at the base of the coxa of the third pair of walking legs and the crabs in the groups 2 and 3 were sacrificed on day 30. After sacrifice, the crabs were weighed and the ovaries were dissected out from the animals and the ovarian indexes were determined using the following formula:

$$\text{Ovarian Index} = \frac{\text{Wet weight of the ovary}}{\text{Wet weight of the crab}} \times 100$$

The ovaries were fixed in aqueous Bouin's fluid. After 24 h fixation, the ovaries were dehydrated through an alcohol series, cleared in xylene and then embedded in

paraffin (melting point 56-58°C). Serial sections (7 μ m) were cut and stained with haematoxylin and counter stained with eosin. The sections were photographed using phase contrast microscope. The data were analyzed using ANOVA followed by Dunnett's test to determine the level of significance.

RESULTS

In *Ozotelpusa senex senex* the immature ovary is translucent to opaque white in color. During early vitellogenesis (Stage I) the gonad color changes from white to yellow. At this stage the yolk globules are deposited in oocytes at the periphery and the oocyte contains prominent nucleus and nucleolus. During late vitellogenesis (Stage II), peripherally located yolk globules moves towards the centre, gradually replacing protein yolk. At this stage, the color of the gonad becomes light orange red. During oogenesis, the oocyte contains a large accumulation of yolk globules occupying the whole of the oocyte and the color of the ovary becomes dark brown or bright orange.

The ovaries were white and the average ovarian index was 0.512 in initial control crabs. The mean ovarian index of concurrent control crabs was 0.514 which is 0.36% more from the initial control group (Table 1) and morphologically ovaries are white in color (Photograph 1A), whereas the mean ovarian index of the leucine

Table 1: Effect of injection of leucine enkephalin on color of the ovary and ovarian index of the freshwater rice field crab, *Ozotelpusa senex senex* Fabricius

Group	Color of the ovary	Ovarian index
Initial control	White	0.512 ^a ±0.008
Concurrent control	White	0.514 ^a ±0.010 (0.39%)
Leucine enkephalin injected	Dark brown/bright orange	1.137±0.025 (122.07%)

P < 0.001

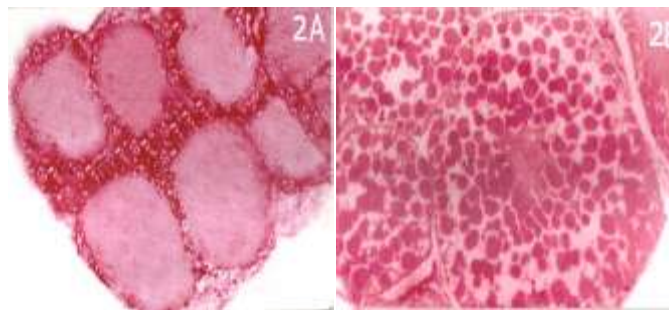
Values are mean ± SD (n=25)

Values in parenthesis are percent change from initial control

Values with the same superscript (a) are not significant



Photograph 1: A) Female reproductive system of concurrent control crab showing immature (White) ovary
B) Female reproductive system of leucine enkephalin injected crab showing fully matured (Brown color) ovary



Photograph 2: A) Transverse section of ovary from concurrent control crab showing previtellogenic oocytes surrounded with thick ovarian wall. Scale line = 4 μ m.
B) Transverse section of ovary from leucine enkephalin injected crab showing lipovitellin yolk globules occupying the whole oocytes. Scale line = 10 μ m.

enkephalin injected crabs were significantly ($p < 0.001$) higher (1.137) with an increased percent of 122.03 than the corresponding values of the initial control crabs. The crabs in group III had dark brown to bright orange color ovary (Table 1; Photograph 1B). Histological section of the ovary from the concurrent control crabs had a thick ovarian wall with a centrally located germanium surrounded by a number of oocytes (Fig. 2A). Whereas oocytes from crabs treated with leucine enkephalin contain a large accumulation of yolk globules occupying the whole of the oocyte (Fig. 2B).

DISCUSSION

In crustaceans, maturation of the ovary can be identified based upon morphological characteristics and coloration. Increase in size as the oocyte proliferation, increase in diameter due to yolk accumulation and change in color from opaque white to dark brown or orange red due to deposition of carotinoids were used as criteria to determine ovarian maturation by several scientists [11]. The concentration of vitellogenin in the hemolymph and an accumulation of yolk globules in oocytes was also used as criteria to determine the ovarian maturation [12]. Recently measurement of ovarian vitellogenin levels at different stages of vary was also used to determine the reproduction with anti vitellin anti-sera [13]. The present study focused on the effect of leucine enkephalin on ovarian maturation in female fresh water rice field crab *Oziotelphusa senex senex* measuring the weight of the ovary in relation to the body weight (ovarian index), color of the ovary and also histological change in the deposition of yolk in oocytes.

The comparison of the present data raised for the initial control crabs and the concurrent control crabs

indicating that the ovaries are developing slowly but not significantly during the course of experimentation. Injection of leucine enkephalin in to female crabs resulted in significant increase in ovarian index and significant histological change in the deposition of yolk in the ovaries in comparison with concurrent control ovarian histology, indicating the involvement of leucine enkephalin in regulating reproduction in crustaceans.

Evidence for a role played by hormones in regulating crustacean reproduction is accumulating. In decapoda crustaceans, ovarian development is under the control of two neuropeptide hormones known as the gonad-inhibiting hormone (GIH) which is synthesized and secreted from medulla terminalis X-organ sinus gland complex present in the eyestalks [14, 15] and the gonad stimulating hormone (GSH) secreted from the brain and thoracic ganglia [16-20]. Besides these, methyl farnesoate (MF) secreted by mandibular organs is also involved in the stimulation of ovarian development [21-29]. Recent studies also indicate that different biogenic amines are also involved in the regulation of crustacean reproduction [30-33].

Our previous studies established the presence of gonad inhibiting hormone, gonad stimulating hormone and methyl farnesoate in *Oziotelphusa senex senex* [24, 27]. Using immunodiagnostic studies it was demonstrated that the presence of leucine enkephalin in the neuroendocrine complex of the eyestalks of several crustacean [2]. Enkephalins mediated release of neurohormones in crustaceans was also established. It was found that leucine enkephalin induces hypoglycaemia in intact *Uca pugilator* but not in eyestalk less individuals. The contrast activity of leucine enkephalin was hypothesized and proved in the red swamp crayfish *Procambarus clarkii* [34]. In view of the

hypothesis proposed experiments was performed and found the inhibitory action of leucine enkephalin on the release of neurohormones(s) from eyestalks.

With the results of the present study we suggest that the induced ovarian growth in the female crab by the leucine enkephalin might be due to inhibition of release of the gonad inhibiting hormone from eyestalks. Further experiments are in progress in this laboratory to firmly establish the role of leucine enkephalin in inducing reproduction in crab *Ozotelphusa senex senex*, since relatively large amounts of opioid peptides are present in the brain and thoracic ganglia [35, 36], where gonad stimulating hormone is synthesized.

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